Biology Class 11 National Book Foundation FBISE Quality Notes
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Chapter 9: Diversity in Plant Functions

9.1 Nutrition in plants (SLO 1, 2)

- ***Nutrient:*** *any substance that supplies essential elements for growth and metabolism.* ***Nutrition*** *= all processes of using nutrients. Plants are autotrophs: they take inorganic nutrients (water, CO_2, minerals) and make organic compounds.*
- ***Sixteen*** *elements are essential.* ***Nine macronutrients*** *(more than 0.05% of dry weight): carbon, hydrogen, oxygen, nitrogen, phosphorus, potassium, sulphur, calcium, magnesium.* ***Seven micronutrients*** *(less than 0.05%): iron, boron, manganese, copper, molybdenum, chlorine, zinc.*
- ***Hydroponic culture*** *tests whether an element is essential: plants grow on a net over aerated water with measured nutrients; water is analysed to see what is absorbed.*

Nutrient	Role	Deficiency
C, H, O	Major components of organic compounds	Poor growth (O: wilting)
Nitrogen (nitrate, ammonium)	Nucleic acids, proteins, chlorophyll	Chlorosis at tips of older leaves
Potassium	Cofactor of enzymes; water balance; stomatal operation	Mottling of older leaves, weak stems, poor roots
Calcium	Middle lamella and cell wall; membrane function; signalling	Crinkling of young leaves, death of terminal buds
Magnesium	Component of chlorophyll; enzyme cofactor	Interveinal chlorosis of older leaves

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Nutrient	Role	Deficiency
Phosphorus	Nucleic acids, phospholipids, ATP	Slow development, thin stems, purpling of veins, poor flowering
Sulphur	Component of proteins	General chlorosis of young leaves
Chlorine	Photosynthesis (water splitting); water balance	Wilting, stubby roots, leaf mottling
Iron	Respiration, photosynthesis, chlorophyll synthesis, N_2 fixation	Interveinal chlorosis of young leaves
Manganese	Amino acid formation; enzyme activation; water splitting	Interveinal chlorosis of young leaves
Boron	Chlorophyll synthesis, cell wall function, pollen tube growth	Death of meristems; thick, leathery leaves
Zinc	Chlorophyll formation; enzyme cofactor; DNA transcription	Reduced internodes, crinkled leaves
Copper	Redox and lignin-biosynthetic enzymes	Light green young leaves, dry tips, stunted branched roots
Molybdenum	Nitrogen metabolism (standard)	General chlorosis, death of leaf tips

- Soil nutrients become deficient when one crop is grown again and again: remedies are **crop rotation** and **fertilizers**.
- **Chlorosis** = yellowing from lack of chlorophyll (deficiency of N, Mg, Fe). **Necrosis** = death of part of the plant (Ca, Mg, Cu, Mo). **Stunted growth** = P, S, K deficiency.

Carnivorous plants (SLO 2)

- Photosynthetic autotrophs that also trap and digest insects as a supplementary **nitrogen** source in nitrogen-poor habitats such as marshes. Modified leaves act as traps; some have symbiotic bacteria that break down insect proteins.

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- *Examples:* ***Pitcher plant*** *(Nepenthes),* ***Venus flytrap*** *(Dionaea muscipula),* ***Sundew*** *(Drosera).*

9.2 Gaseous exchange in plants (SLO 3, 4)

- *Exchange of CO_2 and O_2 between plant and environment is by* ***diffusion*** *through* ***stomata*** *and* ***lenticels****; root epidermis exchanges gases with soil air. Day: CO_2 in, O_2 out (photosynthesis faster). Night: O_2 in, CO_2 out (respiration only). At dawn and dusk there is no net exchange =* ***compensation point****.*
- ***Mesophyll:*** *thin-walled living cells between epidermal layers. Dicots:* ***palisade*** *(tightly packed elongated cells, no spaces, below upper epidermis) and* ***spongy*** *(loose with large air spaces, above lower epidermis) =* ***bifacial / dorsiventral*** *leaf. Monocots: only spongy mesophyll =* ***monofacial / isobilateral****.*

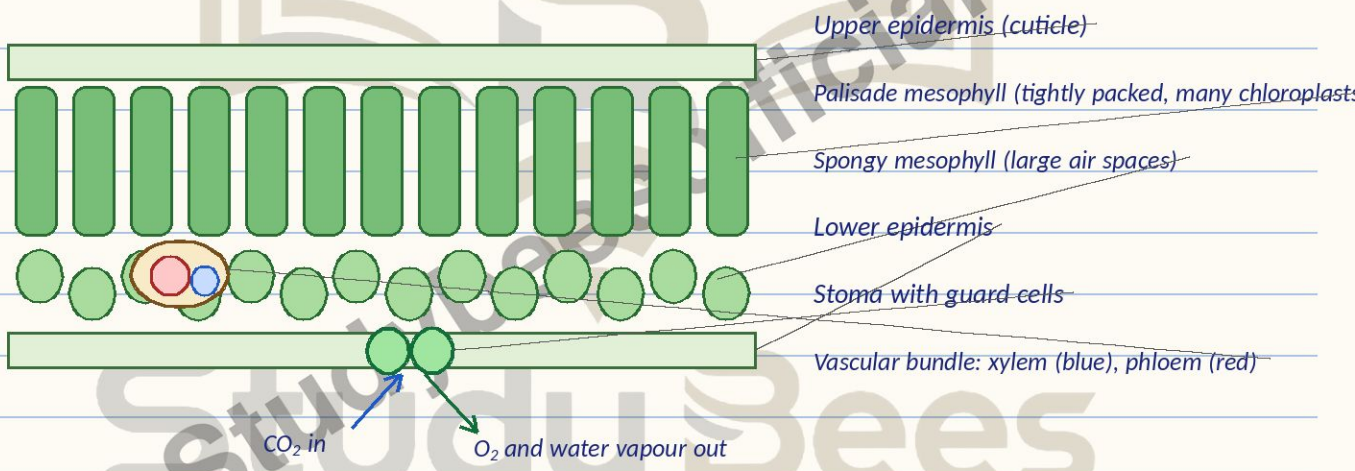


Fig: Bifacial (dorsiventral) leaf - T.S.

- ***Stoma:*** *opening between two kidney (bean) shaped* ***guard cells****. Bifacial leaves have more stomata on the lower epidermis; isobilateral leaves on both equally.*
- ***Transpiration*** *= loss of water vapour from aerial parts mainly through stomata. It moves water and minerals and, since stomata are wide open in daylight for CO_2, links water loss with gas exchange: opening stomata for CO_2 uptake unavoidably loses water vapour.*

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9.3 Transport in plants (SLO 5-10)

Water potential (SLO 6)

- ***Osmosis:*** *movement of water from higher to lower water concentration through a semipermeable membrane. Into cell =* ***endosmosis****; out =* ***exosmosis****.*

Water potential (Psi_w): *Measure of the potential energy of water enabling it to move from one place to another. Unit: megapascal (MPa). Pure water = 0 (highest); all solutions and cells are negative.*

- ***$Psi_w = Psi_s + Psi_p$. Solute (osmotic) potential Psi_s:*** *fall in Psi_w due to dissolved solutes; negative in cells (cytoplasm -0.5 to -1.0 MPa), zero in distilled water.* ***Pressure potential Psi_p:*** *rise in Psi_w due to pressure above atmospheric (turgor).*
- *Water moves from higher to lower Psi_w until equilibrium. For transpiration:* ***Psi_w of soil > root > stem > leaf > atmosphere.***
- ***Worked example:*** *Cell A: Psi_s -2000, Psi_p 600, so Psi_w = -1400 kPa. Cell B: Psi_s -1400, Psi_p 800, so Psi_w = -600 kPa. Cell B has the higher Psi_w, so water moves from B to A by osmosis until the two Psi_w are equal.*

Uptake of water by roots and pathways (SLO 7)

- *Root epidermal cell walls are freely permeable; membrane is differentially permeable. Water enters root hairs by osmosis, passes cortex,* ***endodermis****,* ***pericycle*** *to* ***xylem****. Three pathways:*

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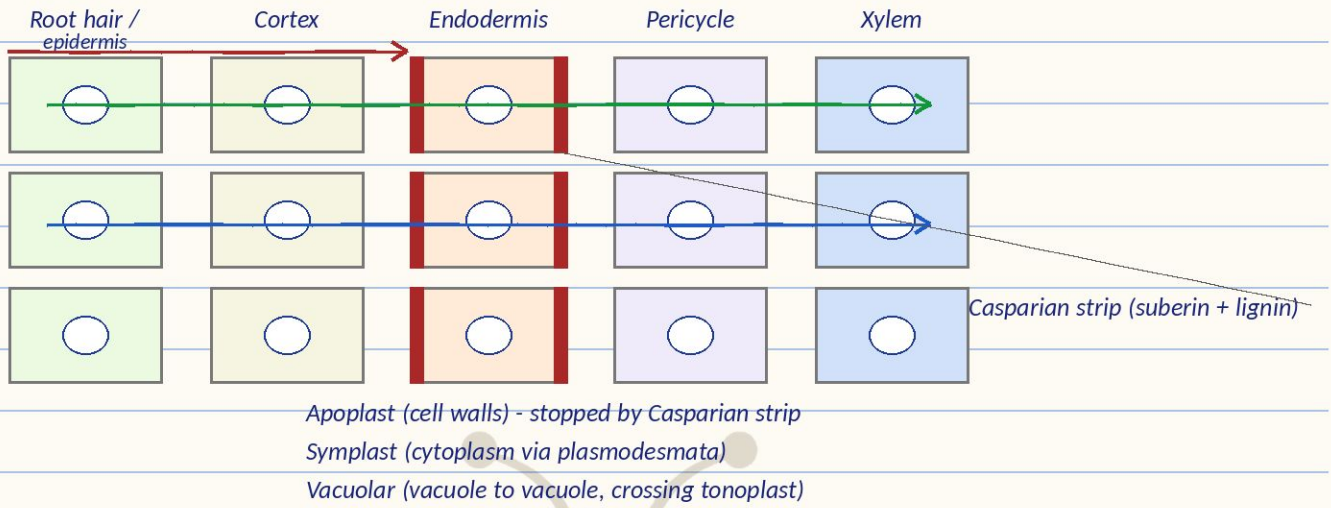


Fig: Pathways of water across the root

1. ***Apoplast:*** *through the continuous system of cell walls and spaces. At the endodermis the* ***Casparian strip*** *(suberin and lignin band on four sides of cells) blocks it; water and ions must cross the membrane into the cell, so the endodermis controls what reaches the xylem.*
2. ***Symplast:*** *through the continuous cytoplasm of cells linked by* ***plasmodesmata****, without crossing further membranes; aided by cytoplasmic streaming.*
3. ***Vacuolar:*** *from vacuole to vacuole, crossing cytoplasm, walls and tonoplast by osmosis down a water potential gradient.*

Structure and function of xylem and phloem (SLO 5)

- ***Xylem:*** *complex permanent tissue; carries water and minerals from roots upward. Four cell types:*

Xylem cell	***Structure and function***
Tracheids	*Dead, elongated, tapering ends, lignified walls; in all vascular plants (tracheophytes); conduct water and give support.*
Vessel elements	*Dead, shorter than tracheids but end-to-end fusion gives long pipelike* ***vessels*** *with thick walls; only in angiosperms; conduct water and support.*

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Xylem cell	Structure and function
Xylem fibres	Dead, narrow, long, thick-walled; mechanical support.
Xylem parenchyma	Only living cells; thin walled; temporary storage of water and minerals.

- **Phloem:** complex permanent tissue; carries organic food (sucrose) up and down. Three living cells and one dead:

Phloem cell	Structure and function
Sieve tube elements	Long tubes by end-to-end fusion; thin walled, living but **no nucleus**; narrow in centre, wide at ends; **sieve plate** (pores) separates successive elements; translocate sucrose.
Companion cells	Living, with nucleus; control the sieve tube elements and supply energy (functional unit with sieve tube).
Phloem parenchyma	Storage cells.
Phloem fibres	Dead, thick walled; mechanical support.

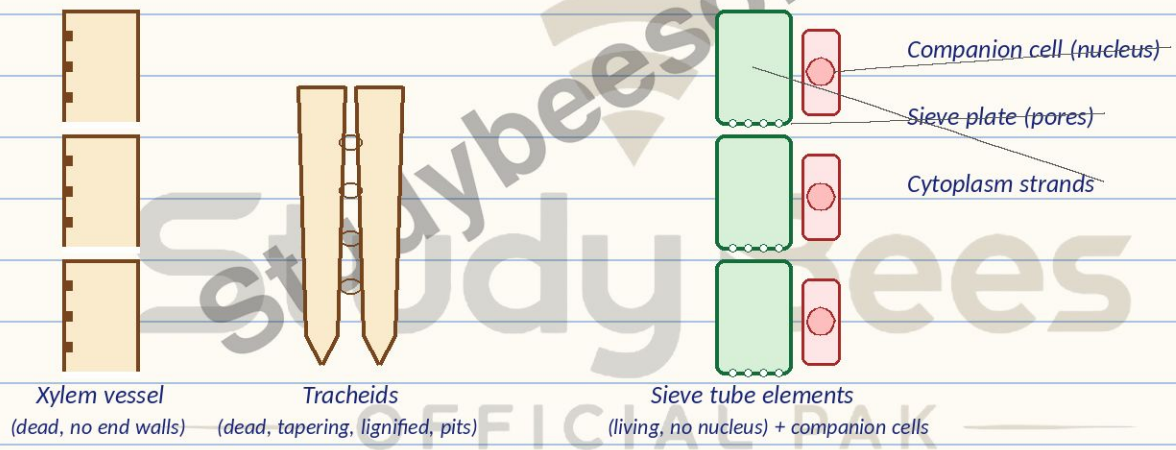


Fig: Xylem vessel, tracheids and phloem sieve tube

Ascent of sap - TACT theory (SLO 8)

- Upward movement of water and minerals from root to leaves through xylem, against gravity. Best theory: **TACT** = **T**ranspiration pull, **A**dhesion, **C**ohesion, **T**ension.

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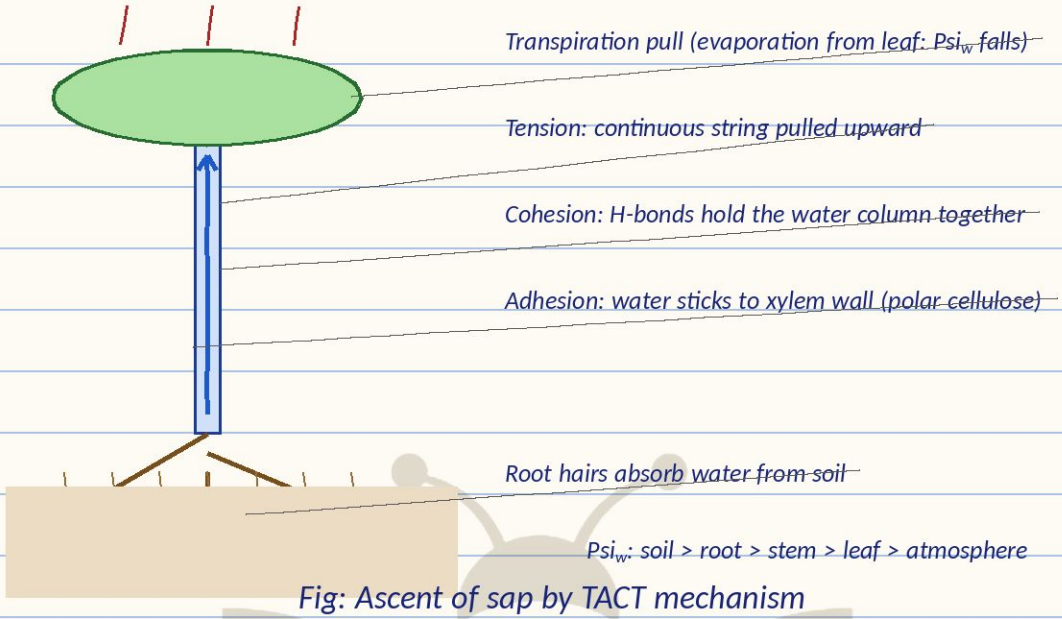


Fig: Ascent of sap by TACT mechanism

1. ***Transpiration pull:*** *evaporation from open stomata makes dry intercellular spaces pull water from mesophyll cells, which pull water from xylem: a pulling force.*
2. ***Adhesion:*** *attraction between water molecules and the polar cellulose walls of xylem capillaries, countering gravity; holds water when transpiration stops.*
3. ***Cohesion:*** *attraction among polar water molecules by hydrogen bonds.*
4. ***Tension:*** *the H-bonds keep the water column continuous and under tension. The column is as strong as a steel wire of the same diameter; as long as transpiration continues it is pulled up as one string, driven by solar energy.*

Opening and closing of stomata (SLO 9)

- *Depends on turgidity of guard cells:* ***turgid = open, flaccid = closed.*** *Open by day, partly closed at night.*

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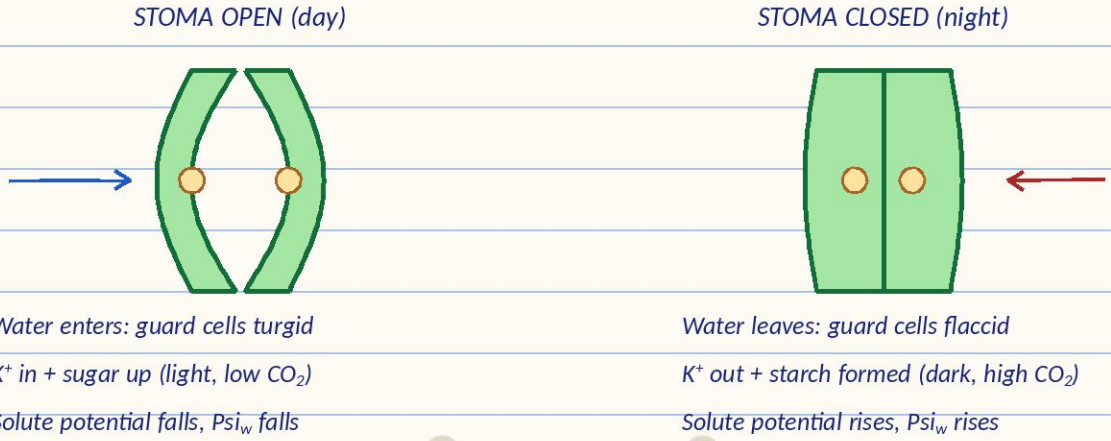


Fig: Opening and closing of stomata

- ***Starch-sugar interconversion theory:*** *guard cells have chloroplasts. In light photosynthesis raises glucose, so solute potential and Psi_w fall (more negative); water enters from neighbouring epidermal cells; guard cells turgid, stoma opens. In dark glucose is used up, converted to sucrose and starch; Psi_w rises; water leaves; guard cells flaccid, stoma closes.*
- ***K^+ influx / efflux theory:*** *in light CO_2 in guard cells falls; K^+ actively enter from epidermal cells; malic acid ionizes (blue light) to malate and H^+; H^+ pumped out; K^+ + malate = potassium malate lowers Psi_w: water enters, stoma opens. At night CO_2 rises, K^+ leave by efflux, malic acid reforms; Psi_w rises; water leaves, stoma closes.*

Translocation of organic solutes (SLO 10)

- *Movement of prepared food through phloem. Direction is **source to sink**, not simply up or down. **Source** = supply area (leaf: purely a source; storage organ when it supplies). **Sink** = area of use or storage (fruit: purely a sink; root and stem both).*
- ***Phloem sap:*** *10-25% dry matter, of which about 90% is **sucrose**.*

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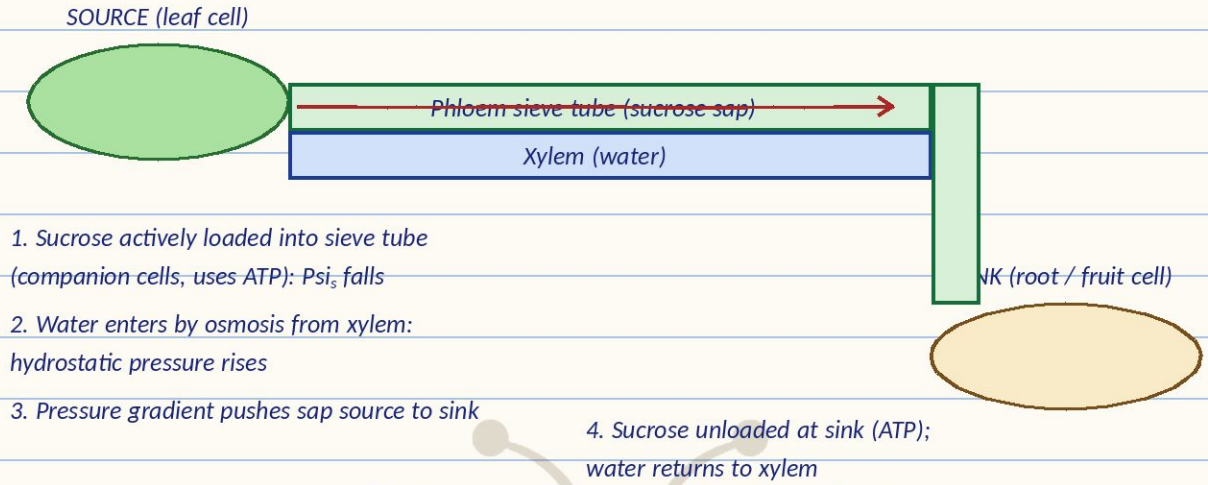


Fig: Pressure flow (mass flow) hypothesis

- ***Pressure flow (mass flow) theory:*** *sugars made at source are actively loaded (companion cells, ATP) into sieve tubes, lowering Psi$_s$ and Psi$_w$; water enters by osmosis from xylem and raises hydrostatic pressure, which pushes sap away from the source. At the sink sugars are actively unloaded, Psi$_w$ rises and water returns to xylem. The pressure gradient from high (source) to low (sink) drives translocation.*

9.4 Homeostasis in plants (SLO 11-14)

Osmotic adjustment (osmoregulation): *Maintenance of water and solute levels in the body.*

Medium	Water potential / solute	Water movement and cell
Hypotonic	Psi$_w$ higher than cell sap; solute lower	Endosmosis: water in; cell turgid
Hypertonic	Psi$_w$ lower than cell sap; solute higher	Exosmosis: water out; cell flaccid (plasmolysis)
Isotonic	Same as cell sap	No net movement (ideal; rare in nature)

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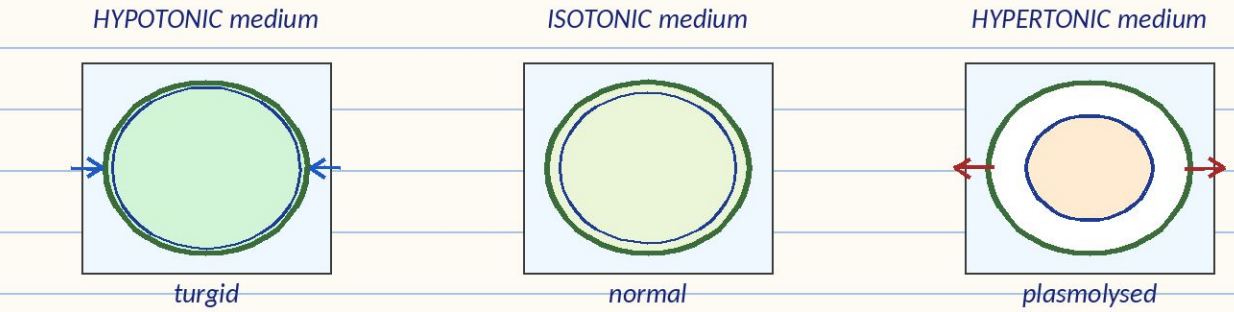


Fig: Cell in hypotonic, isotonic and hypertonic media

Osmotic adjustment in plants of different habitats (SLO 13)

Feature	Hydrophytes (aquatic)	Mesophytes (moderate)	Xerophytes (dry)
Medium	Hypotonic: flooding	Nearly isotonic	Strongly hypertonic: dehydration
Transpiration	Very high	Medium	Very low
Leaves	Broad, large surface	Medium	Needle-like or spines
Stomata	Upper epidermis; open day and night	Lower epidermis; open day, close night	Sunken in depressions; close day, open night
Cuticle	Very thin or none	Thin	Very thick
Water storage	None	Little	High (succulent)
Examples	Water lily, Hydrilla	Wheat, rice, maize	Cactus, Opuntia

- ***Halophytes (saline soil):*** *high salt gives very low Psi_w:* ***physiological drought*** *(plants cannot absorb water though it is present). Halophytes either take up Na^+ and Cl^- at high rates and store them in leaf* ***vacuoles*** *(keeping cytoplasm low in salt) to adjust osmotically, or accumulate* ***compatible solutes*** *(organic: glycine, betaine, sugar alcohols like sorbitol) that do not disturb enzymes.*

Thermoregulation (SLO 14)

- ***Low temperature:*** *membrane lipids crystallize, becoming porous; protein structure is affected; ice crystals form in cytoplasm. Adaptations: more* ***unsaturated fatty acids*** *(low freezing point) in membranes; changed solute composition so cytosol* ***supercools*** *without ice (ice may form in cell wall).*

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Gradual chilling is less stressful than rapid chilling.

- ***High temperature:*** *excess transpiration, dehydration, wilting, enzyme denaturation. Adaptations: shiny reflective cuticle; many stomata to lose heat by transpiration (but above 45 C stomata close); synthesis of* ***heat shock proteins*** *that protect enzymes from denaturation.*

9.5 Support in plants (SLO 15, 16)

- *Mechanisms: turgor pressure in parenchyma, mechanical tissues (collenchyma, sclerenchyma), arrangement of vascular bundles, secondary growth.*
- ***Turgor pressure:*** *parenchyma cells with large water-filled vacuoles press on walls and keep cells stiff. In herbaceous plants turgidity gives support and upright growth; loss of turgor (hydrostatic pressure) causes wilting.*

Collenchyma	***Sclerenchyma***
Living; extra cellulose at corners	*Dead at maturity; thick lignified secondary wall*
Supports young plants, herbs, leaves (no secondary growth); can grow and stretch	*Solely for support and strength; great tensile strength*
In stems and petioles near epidermis (peripheral)	*Fibres (elongated; pericycle, xylem, phloem) and sclereids (spherical; fruit wall, seed coats)*

- ***Vascular bundles:*** *ring in dicot stems (also supports); scattered in monocots.*

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9.6 Growth and development (SLO 17-19)

Growth: *Irreversible increase in size and mass of an organism. Development = programmed series of changes from simple to more complex form.*

- *Four phases: cell division, cell elongation, maturation and differentiation. Plants keep adding organs throughout life =* ***open growth****.*
- ***Meristems:*** *growing regions of small, cuboidal cells with large nuclei, small vacuoles and thin walls. Division gives an* ***initial*** *(stays in meristem) and a* ***derivative*** *(differentiates).*

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Apical meristem	Lateral meristem
Tips of roots and shoots	Cylinders on sides of stem and root (dicots, gymnosperms)
Primary meristem (from embryo); lengthens plant	Secondary meristem; increases thickness
Primary growth (indeterminate)	Secondary growth: vascular cambium, cork cambium; intercalary meristem at nodes (determinate)

- ***Primary growth:*** *increase in length by apical meristem.* ***Secondary growth:*** *increase in thickness by lateral meristems; tissues formed are secondary tissues.*
- ***Vascular cambium:*** *between primary xylem and phloem; adds* ***secondary xylem*** *inward (faster) and* ***secondary phloem*** *outward.* ***Cork cambium (phellogen):*** *from outer cortex; adds* ***secondary cortex*** *inward and* ***cork*** *outward; cork cells die so mature bark is rough.*

Annual rings (SLO 18)

- *In spring and summer (water available) vascular cambium makes large cells; in autumn and winter (water scarce) small cells. The alternating rings of larger and smaller cells = one* ***growth (annual) ring*** *per year; counting rings gives age.*

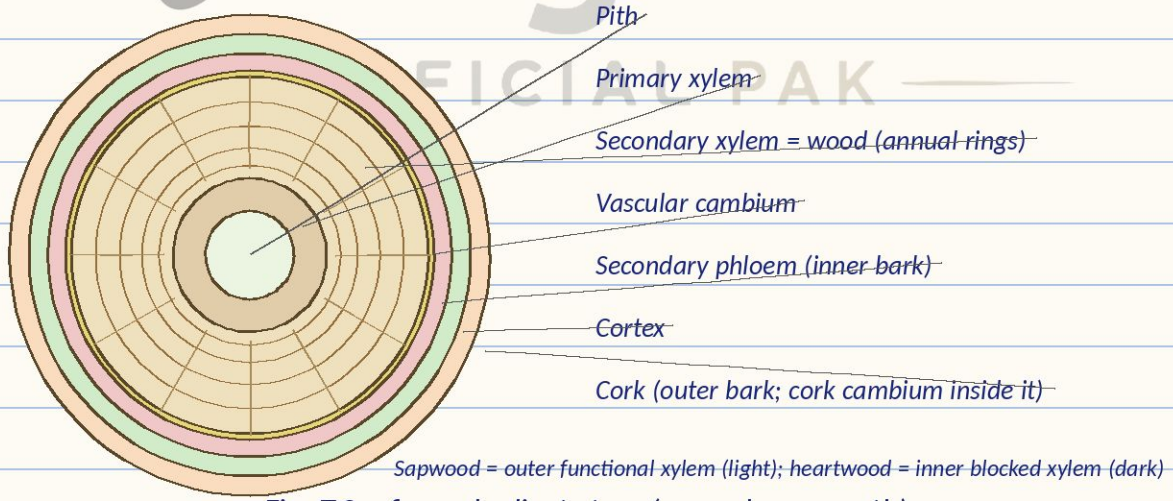


Fig: T.S. of woody dicot stem (secondary growth)

- ***Wood*** *= secondary xylem.* ***Heartwood*** *= inner, dark, blocked with wastes.*

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Sapwood *= outer, light, functional.* ***Bark*** *= everything outside the vascular cambium: inner bark (secondary phloem, primary phloem, pericycle, endodermis, cortex) and outer bark (cork and epidermis).*

- *Tropical trees often lack distinct rings because there is no seasonal change (growth is continuous). Hardwood is denser, stronger and more durable than softwood, hence preferred for furniture.*

Growth correlations and apical dominance (SLO 19)

- *Interactions among growing points.* ***Apical dominance*** *= negative correlation: the growing apical bud inhibits lateral buds (and the main root inhibits lateral roots). Caused by* ***auxin*** *from the apical bud;* ***cytokinin*** *from lateral buds works antagonistically. Strong dominance: tall, short side branches; weak: short, dense lateral branches.*

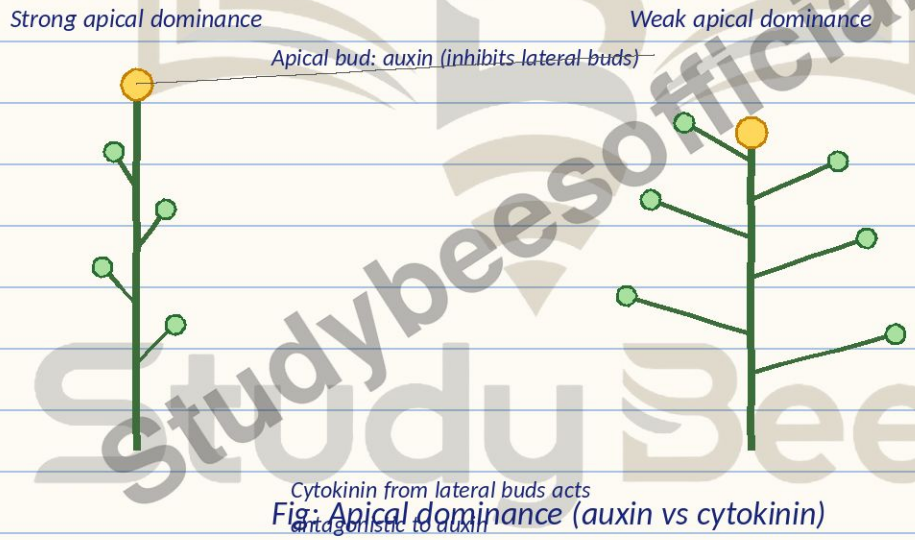


Fig: Apical dominance (auxin vs cytokinin)

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9.7 Growth responses in plants (SLO 20-24)

Plant growth regulators (SLO 20)

- *Five major groups: auxins, gibberellins, cytokinins (promoters), abscisic acid and ethene (inhibitors).*

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Regulator	Source and effects
Auxin (IAA)	First hormone discovered; made in shoot tips, leaf primordia, young leaves, embryos, flowers and seeds; promotes **cell elongation** (stem elongation, leaf expansion); apical dominance; root formation.
Gibberellins	More than 75 (GA_1, GA_3...); abundant in seeds and young shoots; stem elongation by cell division and elongation; seed germination (break dormancy); also vernalin.
Cytokinins	Similar to adenine; zeatin (from corn); in root tips, seeds, fruits, leaves; promote cell division; delay senescence; lateral bud growth. High cytokinin : auxin forms shoots; low forms roots.
Abscisic acid (ABA)	Made in plastids from carotenoids under stress; inhibits cell growth; promotes seed dormancy; water stress raises ABA and closes stomata.
Ethene (ethylene)	Gaseous hydrocarbon from an amino acid, rises under stress; stimulates fruit ripening and abscission of fruits and leaves.

- Hormones for: germination of seeds = gibberellins; stem elongation = auxin and gibberellins; ripening of fruits = ethene; abscission of leaves = ethene (and ABA); dormancy of seeds = abscisic acid.

Tropic movements (SLO 21)

Tropism	Stimulus and response
Phototropism	Light. Shoots positive (towards light); roots negative.
Geotropism (gravitropism)	Gravity. Roots positive (down); shoots negative (up).
Thigmotropism	Touch. Climbing plants coil around wall, stick, rope.
Chemotropism	Chemicals / nutrients. Fungal hyphae grow toward nutrients (pollen tube towards ovule).

Photoperiodism (SLO 22, 23)

Photoperiodism: Effect of the total duration of light in a day (photoperiod) on development of flowers. Appropriate photoperiod turns vegetative buds into floral buds.

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Type	Response and examples
Short-day plants	Flower when photoperiod is less than a critical length: Maryland mammoth tobacco, cocklebur (critical 15.5 h), chrysanthemum, tobacco (14 h).
Long-day plants	Flower when photoperiod exceeds the critical length: spinach, sugar beet, henbane (11 h), snapdragon, cabbage, spring wheat.
Day-neutral plants	Flower in any photoperiod after enough vegetative growth: tomato, pansy, bean, sweet pea, rose.

- ***Critical photoperiod:*** *for SDP the maximum day length that induces flowering; for LDP the minimum.*
- ***The dark period is what matters:*** *uninterrupted dark is the real stimulus. Short-day plants are really* ***long-night plants****; long-day plants are* ***short-night plants****. Interrupting the night of cocklebur with light prevents flowering; henbane flowers in short days if its long night is interrupted.*
- ***Light quality:*** *red light (660 nm) in a night break prevents flowering in SDP; far-red (730 nm) reverses it; the last exposure determines the response. This led to the discovery of* ***phytochromes****.*

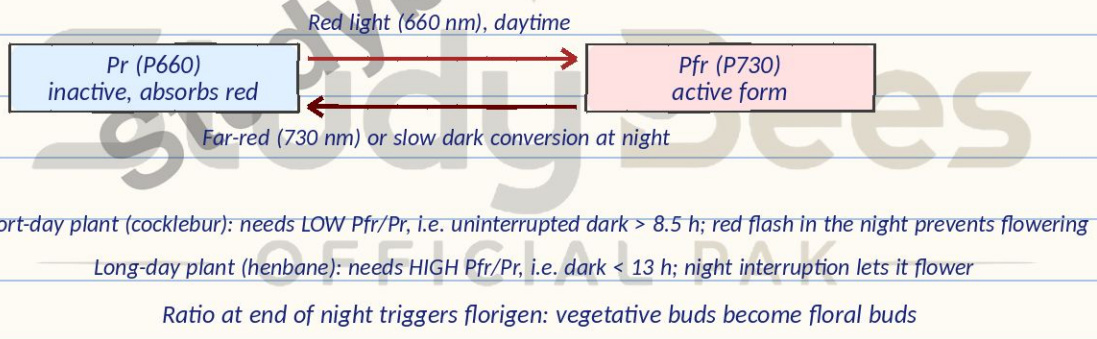


Fig: Phytochrome interconversion and photoperiodism

- ***Phytochromes:*** *blue-green protein pigments with an iron-containing non-protein part. Two interconvertible forms:* ***Pr (P660)*** *absorbs red and converts to* ***Pfr (P730)*** *(daytime); Pfr absorbs far-red (or slowly in dark) and reverts to Pr (mainly at night). The conversion rate acts as a biological clock to measure night length.*
- ***SDP*** *needs a* ***low Pfr : Pr*** *ratio: with 8.5 h or longer of dark, all Pfr converts*

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*back. **LDP** needs a **high Pfr : Pr** ratio: with dark of 13 h or less, Pfr is not all converted. The ratio at the end of night triggers **florigen**, which converts vegetative buds into floral buds.*

Vernalization (SLO 24)

***Vernalization:** Promotion of flowering by exposure to low temperature.*

- *Received by the **shoot apex** of a mature plant or embryo (not leaves). Usually 0-10 C, most effective about 4 C. Needed by **biennials**, e.g. winter wheat sown in October-November and harvested in June-July. Spring-sown winter wheat does not flower unless seeds are chilled (4 C for several weeks).*
- *Low temperature stimulates **vernalin**, which turns vegetative buds into floral buds; vernalin is actually **gibberellin**. In some plants vernalization is an absolute requirement.*

STEAM activity: plant vascular systems

- *Place celery stems in red / blue coloured water, watch dye move up the xylem, cut cross sections and view under a microscope to identify xylem and phloem; discuss capillary action and how temperature or light affect water transport.*

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Chapter 9: Short Questions with Answers (SLO based)

Q1. [SLO 1] Define nutrient and nutrition.

Ans: A nutrient supplies essential elements for growth and metabolism; nutrition is all the processes of utilizing nutrients.

Q2. [SLO 1] What are macronutrients and micronutrients? Name them.

Ans: Macronutrients (above 0.05% of dry weight): C, H, O, N, P, K, S, Ca, Mg. Micronutrients (below 0.05%): Fe, B, Mn, Cu, Mo, Cl, Zn.

Q3. [SLO 1] How many elements are essential for plants?

Ans: Sixteen: nine macronutrients and seven micronutrients.

Q4. [SLO 1] What is hydroponic culture?

Ans: A method to test whether an element is essential: plants grow on a net over aerated water containing measured nutrients, and the water is analysed for what is absorbed.

Q5. [SLO 1] What is the role of nitrogen? What does its deficiency cause?

Ans: Component of nucleic acids, proteins and chlorophyll; deficiency causes chlorosis at tips of older leaves.

Q6. [SLO 1] Why is magnesium important?

Ans: It is a component of chlorophyll and an enzyme cofactor; deficiency causes interveinal chlorosis in older leaves.

Q7. [SLO 1] Write the role of calcium and potassium.

Ans: Calcium: middle lamella, cell wall, membrane function. Potassium: enzyme cofactor, water balance, stomatal operation.

Q8. [SLO 1] What are the roles of phosphorus and sulphur?

Ans: Phosphorus: nucleic acids, phospholipids, ATP. Sulphur: component of proteins.

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Q9. [SLO 1] Differentiate chlorosis, necrosis and stunted growth.

Ans: Chlorosis: yellowing due to lack of chlorophyll. Necrosis: death of part of the plant. Stunted growth: less than normal growth (deficiency of P, S, K).

Q10. [SLO 1] How can nutrient deficiency in soil be remedied?

Ans: By crop rotation and by adding fertilizers.

Q11. [SLO 2] What are carnivorous plants? Give examples.

Ans: Photosynthetic plants that trap and digest insects for nitrogen: pitcher plant, Venus flytrap, sundew.

Q12. [SLO 2] Why do carnivorous plants need insects?

Ans: They grow in nitrogen-poor marshy soils; insects are a supplementary source of nitrogen.

Q13. [SLO 3] What are the openings for gaseous exchange in plants?

Ans: Stomata and lenticels.

Q14. [SLO 3] What is the compensation point?

Ans: The point at dawn and dusk where photosynthesis equals respiration so there is no net gas exchange.

Q15. [SLO 3] Differentiate between palisade and spongy mesophyll.

Ans: Palisade: tightly packed elongated cells under upper epidermis, no spaces. Spongy: loosely arranged cells with large air spaces near lower epidermis.

Q16. [SLO 3] What are bifacial and isobilateral leaves?

Ans: Bifacial (dicot): palisade and spongy mesophyll, distinct upper and lower surfaces. Isobilateral (monocot): only spongy mesophyll, similar surfaces.

Q17. [SLO 3] Why is gas exchange more efficient in air than in water?

Ans: Gases diffuse far faster in air and air holds much more oxygen than water.

Q18. [SLO 4] Define transpiration. Relate it to gas exchange.

Ans: Loss of water vapour from aerial parts mainly through stomata. Stomata open for CO_2 uptake, so transpiration unavoidably goes with gas exchange.

Q19. [SLO 4] Differentiate between transpiration and evaporation.

Ans: Transpiration is water loss from living plant parts mainly through stomata and is regulated; evaporation is physical loss from any wet surface.

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Q20. [SLO 5] Name the four types of xylem cells.

Ans: Tracheids, vessel elements, xylem fibres and xylem parenchyma.

Q21. [SLO 5] Which xylem cells are living?

Ans: Only xylem parenchyma.

Q22. [SLO 5] Differentiate between tracheids and vessels.

Ans: Tracheids: single dead tapering lignified cells in all vascular plants. Vessels: tubes formed by end-to-end fusion of vessel elements, only in angiosperms, more efficient.

Q23. [SLO 5] What are sieve tube elements?

Ans: Living thin-walled cells without nuclei, fused end to end, separated by sieve plates; they translocate sucrose.

Q24. [SLO 5] What is the role of companion cells?

Ans: They have a nucleus, control the sieve tube elements and supply energy for their activities.

Q25. [SLO 5] What is a sieve plate?

Ans: A transverse or oblique pore-bearing partition between two successive sieve tube elements.

Q26. [SLO 6] What is water potential? What is its value in pure water?

Ans: The potential energy of water enabling it to move; measured in MPa. Pure water has the highest value, zero; solutions are negative.

Q27. [SLO 6] Write the equation for water potential.

Ans: $Psi_w = Psi_s + Psi_p$ (solute potential plus pressure potential).

Q28. [SLO 6] Define solute potential and pressure potential.

Ans: Solute potential: fall in water potential due to dissolved solutes (negative). Pressure potential: rise in water potential due to pressure above atmospheric (turgor).

Q29. [SLO 6] Why is solute potential also called osmotic potential?

Ans: Because the lower Psi_s of cell cytoplasm makes water enter cells by osmosis.

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Q30. [SLO 6] Define osmosis and diffusion.

Ans: Osmosis: water movement through a semipermeable membrane from high to low water concentration. Diffusion: movement of molecules from high to low concentration.

Q31. [SLO 7] What are apoplast, symplast and vacuolar pathways?

Ans: Apoplast: through cell walls. Symplast: through connected cytoplasm via plasmodesmata. Vacuolar: vacuole to vacuole across tonoplast.

Q32. [SLO 7] What is the Casparian strip?

Ans: A band of suberin and lignin on four sides of endodermal cells that blocks the apoplast pathway so water and ions must enter the cell cytoplasm.

Q33. [SLO 7] How does symplast differ from apoplast?

Ans: Symplast: cytoplasm of cells joined by plasmodesmata. Apoplast: continuous system of cell walls and spaces.

Q34. [SLO 7] What is the path of water and salts in a vascular plant?

Ans: Soil to root hair, epidermis, cortex, endodermis, pericycle, xylem, then up to leaves.

Q35. [SLO 8] What is ascent of sap?

Ans: Upward movement of water and minerals from root to leaves through xylem.

Q36. [SLO 8] What is TACT theory?

Ans: Transpiration pull, Adhesion, Cohesion and Tension act together to lift water in xylem.

Q37. [SLO 8] Define adhesion and cohesion.

Ans: Adhesion: attraction between water and other substances such as xylem walls. Cohesion: attraction among water molecules by hydrogen bonds.

Q38. [SLO 8] What is the source of energy for ascent of sap?

Ans: Solar energy driving evaporation (transpiration) from leaves.

Q39. [SLO 9] What controls the opening of stomata?

Ans: Turgidity of guard cells: turgid = open; flaccid = closed.

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Q40. [SLO 9] State the K^+ theory of stomatal opening.

Ans: In light CO_2 falls, K^+ enter guard cells and form potassium malate, Psi_w falls, water enters, guard cells turgid and the stoma opens.

Q41. [SLO 9] Why do guard cells differ from other epidermal cells?

Ans: They are the only epidermal cells with chloroplasts and so photosynthesize.

Q42. [SLO 10] Define translocation, source and sink.

Ans: Translocation: movement of organic food through phloem. Source: area supplying food (leaf). Sink: area using or storing food (fruit, root).

Q43. [SLO 10] What is the composition of phloem sap?

Ans: 10-25% dry matter, about 90% of which is sucrose.

Q44. [SLO 10] State the pressure flow hypothesis.

Ans: Sugar is actively loaded at the source, water enters by osmosis and raises pressure, pushing sap to the sink where sugar is unloaded and water returns to xylem.

Q45. [SLO 11] Define osmotic adjustment and homeostasis.

Ans: Osmotic adjustment: maintenance of water and solute level in the body. Homeostasis: ability to maintain a steady state in a changing environment.

Q46. [SLO 12] What happens to a cell in hypotonic and hypertonic solution?

Ans: Hypotonic: water enters (endosmosis), cell turgid. Hypertonic: water leaves (exosmosis), cell flaccid.

Q47. [SLO 12] What is an isotonic condition?

Ans: Water potential of cell and environment are equal, so no net movement of water.

Q48. [SLO 13] Why do hydrophytes have the highest rate of transpiration?

Ans: They have broad leaves with large surface area and stomata on the upper epidermis kept open day and night.

Q49. [SLO 13] List adaptations of xerophytes.

Ans: Needle-like leaves or spines, sunken stomata closed in day, thick cuticle, water storage (succulents).

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Q50. [SLO 13] What is physiological drought?

Ans: In saline soil low water potential prevents roots from absorbing water even though water is present.

Q51. [SLO 13] What are halophytes? What are compatible solutes?

Ans: Plants of saline soils. Compatible solutes are organic compounds (glycine, sugar alcohols like sorbitol) accumulated for osmotic adjustment without harming enzymes.

Q52. [SLO 14] How do plants adapt to low temperature?

Ans: More unsaturated fatty acids in membranes and changed solute composition so cytosol supercools without ice.

Q53. [SLO 14] What are heat shock proteins?

Ans: Heat-tolerant proteins made at high temperature that protect enzymes and other proteins from denaturation.

Q54. [SLO 15] What is turgor pressure? Why is it important in herbs?

Ans: Pressure of water-filled vacuoles on cell walls; it keeps herbaceous plants upright; loss of it causes wilting.

Q55. [SLO 16] Why is support needed in terrestrial life?

Ans: Without water buoyancy, plants need support to stay upright against gravity and expose leaves to light.

Q56. [SLO 16] Name the supporting tissues of plants.

Ans: Parenchyma (turgor), collenchyma, sclerenchyma (fibres and sclereids), xylem and phloem.

Q57. [SLO 16] Differentiate between collenchyma and sclerenchyma.

Ans: Collenchyma: living, extra cellulose at corners, supports young organs. Sclerenchyma: dead, lignified, for strong support (fibres and sclereids).

Q58. [SLO 17] Define growth and development.

Ans: Growth: irreversible increase in size and mass. Development: programmed changes from simpler to more complex form.

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Q59. [SLO 17] What are meristems?

Ans: Tissues of small young cells that keep dividing in growth regions (apical and lateral meristems).

Q60. [SLO 17] Differentiate between primary and secondary growth.

Ans: Primary: increase in length by apical meristems. Secondary: increase in thickness by vascular and cork cambium in dicots and gymnosperms.

Q61. [SLO 17] What happens to primary tissues during secondary growth?

Ans: They are pushed inward / outward (primary xylem inside, primary phloem and cortex squeezed outward) as secondary tissues are added.

Q62. [SLO 18] What are annual rings? Why do they form?

Ans: Rings of smaller and larger xylem cells showing one year of growth; formed by seasonal change in water availability.

Q63. [SLO 18] Why does wood of many tropical trees lack annual rings?

Ans: There are no distinct seasons, so cambium works continuously and makes uniform cells.

Q64. [SLO 18] What are heartwood and sapwood?

Ans: Heartwood: dark inner non-functional wood blocked with wastes. Sapwood: light outer functional xylem.

Q65. [SLO 18] Why is hardwood more desirable than softwood for furniture?

Ans: It is denser, stronger and more durable.

Q66. [SLO 19] What is apical dominance?

Ans: Inhibition of lateral buds by the apical bud, caused by auxin from the apical bud.

Q67. [SLO 20] Name the plant growth regulators.

Ans: Auxins, gibberellins, cytokinins (promoters), abscisic acid and ethene (inhibitors).

Q68. [SLO 20] Name the hormones for germination, stem elongation, fruit ripening, leaf abscission and seed dormancy.

Ans: Gibberellins; auxin and gibberellins; ethene; ethene (and ABA); abscisic acid.

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Q69. [SLO 20] What are the roles of auxin and cytokinin?

Ans: Auxin promotes cell elongation and apical dominance; cytokinin promotes cell division and lateral bud growth.

Q70. [SLO 20] How does ABA help in water stress?

Ans: ABA rises in water-stressed leaves and closes stomata.

Q71. [SLO 21] Define phototropism and geotropism.

Ans: Phototropism: growth movement towards or away from light. Geotropism: growth movement towards or away from gravity.

Q72. [SLO 21] What is thigmotropism? Give an example.

Ans: Movement in response to touch, as in climbing plants coiling around a support.

Q73. [SLO 21] What is chemotropism?

Ans: Growth movement in response to a chemical, e.g. fungal hyphae growing toward nutrients.

Q74. [SLO 22] Define photoperiodism. Differentiate it from phototropism.

Ans: Photoperiodism: effect of day length on flowering. Phototropism: bending of organs toward light.

Q75. [SLO 23] Classify plants on the basis of photoperiodism with one example each.

Ans: Short-day (cocklebur), long-day (spinach, henbane), day-neutral (tomato).

Q76. [SLO 23] What is critical photoperiod?

Ans: For a short-day plant, the maximum day length that allows flowering; for a long-day plant, the minimum day length needed.

Q77. [SLO 23] What are phytochromes?

Ans: Photoreceptor pigments (Pr and Pfr forms) that sense red and far-red light and control flowering.

Q78. [SLO 23] What is florigen?

Ans: A flowering hormone produced when the right Pfr : Pr ratio is reached; it turns vegetative buds into floral buds.

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Q79. [SLO 24] Define vernalization.

Ans: Promotion of flowering by exposure to low temperature (about 4 C).

Q80. [SLO 24] Why does winter wheat need vernalization?

Ans: It is a biennial; sown in autumn it receives cold; if sown in spring it will not flower unless seeds are chilled.

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Chapter 9: Long Questions with Answers (SLO based)

Q1. [SLO 1] List the macro and micronutrients in plants highlighting the role of each.

Ans:

- *Macronutrients (9): C, H, O (organic compounds), N (proteins, nucleic acids, chlorophyll), P (nucleic acids, ATP, phospholipids), K (enzymes, stomata, water balance), S (proteins), Ca (cell wall, middle lamella), Mg (chlorophyll).*
- *Micronutrients (7): Fe (chlorophyll synthesis, respiration), B (pollen tube, cell wall), Mn (water splitting, enzymes), Cu (redox enzymes), Mo (nitrogen metabolism), Cl (water splitting), Zn (enzyme cofactor, DNA transcription).*
- *Deficiencies: chlorosis (N, Mg, Fe), necrosis (Ca, Mg, Cu, Mo), stunted growth (P, S, K).*

Q2. [SLO 3] Explain the role of stomata and palisade tissue in the exchange of gases in plants.

Ans:

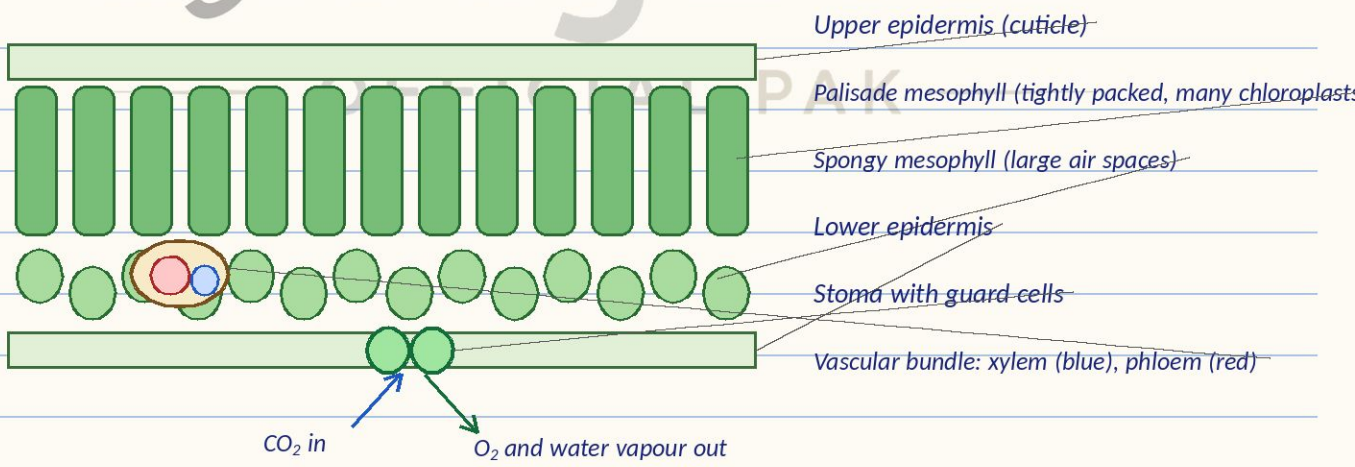


Fig: Bifacial (dorsiventral) leaf - T.S.

- *Palisade mesophyll has tightly packed, chloroplast-rich cells for photosynthesis;*

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spongy mesophyll has air spaces for gas diffusion.

- *Stomata (guard cells) open in day: CO_2 diffuses in through air spaces to palisade cells; O_2 and water vapour diffuse out. At night O_2 enters for respiration.*

Q3. [SLO 4] Relate transpiration with gas exchange in plants.

Ans:

- *Transpiration is water vapour loss through stomata. Stomata must open for CO_2 uptake and O_2 release, so water loss is unavoidable.*
- *It also draws water and minerals upward (transpiration pull) and cools the leaf; stomata are wide open in the day for both gas exchange and transpiration.*

Q4. [SLO 5] Describe the structure of xylem vessel elements, sieve tube elements, companion cells and tracheids and relate structure to function.

Ans:

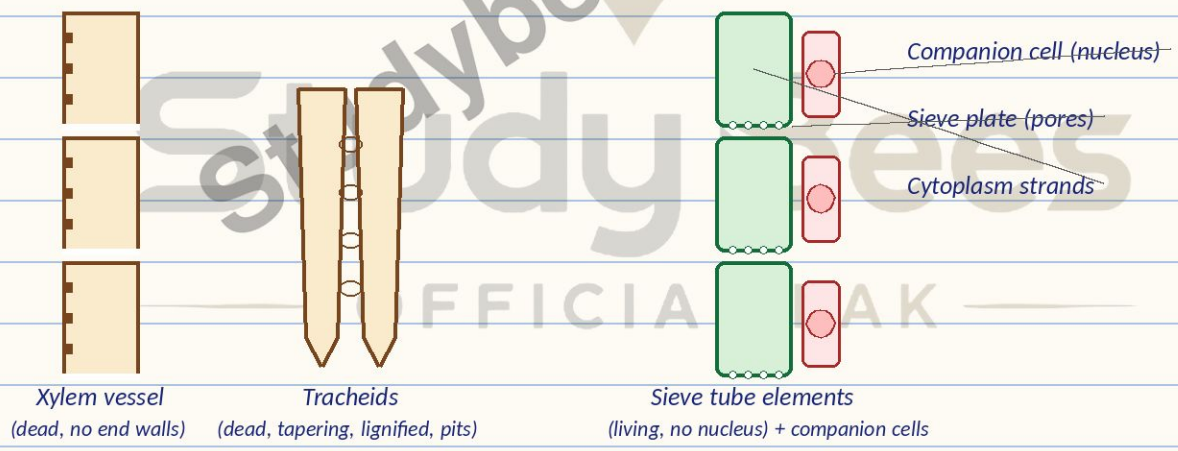


Fig: Xylem vessel, tracheids and phloem sieve tube

- *Tracheids: dead, tapering, lignified, with pits: conduct water and support. Vessel elements: dead, wide, end walls lost so form long pipes: fast water flow, support.*
- *Sieve tube elements: living, no nucleus, sieve plates with pores: sucrose flows through. Companion cells: nucleated, controlling and energizing sieve tubes.*

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Q5. [SLO 6] Explain movement of water between plant cells and between cells and their environment in terms of water potential.

Ans:

- $Psi_w = Psi_s + Psi_p$; pure water 0; solutes lower Psi_s; turgor raises Psi_p. Water moves from higher to lower Psi_w by osmosis until equilibrium.
- Cell sap has a more negative Psi_w than soil water, so water enters root cells; between cells, water moves from the cell with higher to the one with lower Psi_w. Psi_w: soil > root > stem > leaf > atmosphere.

Q6. [SLO 7] Explain movement of water in the root through symplast, apoplast and vacuolar pathways.

Ans:

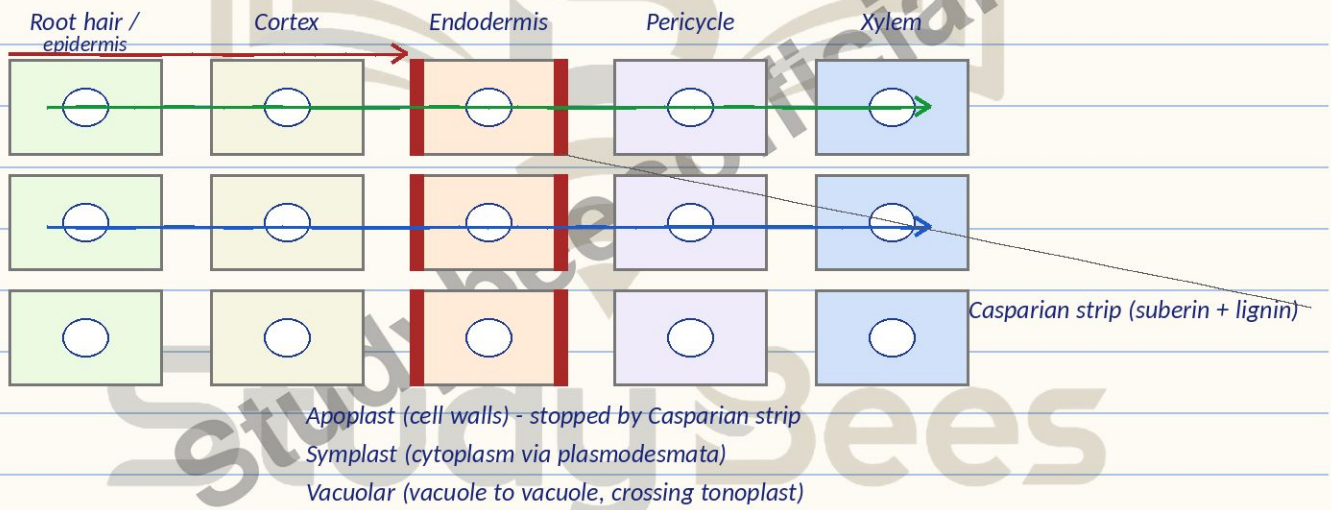


Fig: Pathways of water across the root

- Apoplast: cell walls, stopped at endodermis by Casparian strip. Symplast: cytoplasm via plasmodesmata. Vacuolar: vacuole to vacuole across tonoplast. All finally reach the xylem.

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Q7. [SLO 8] Explain movement of water in xylem through the TACT mechanism.

Ans:

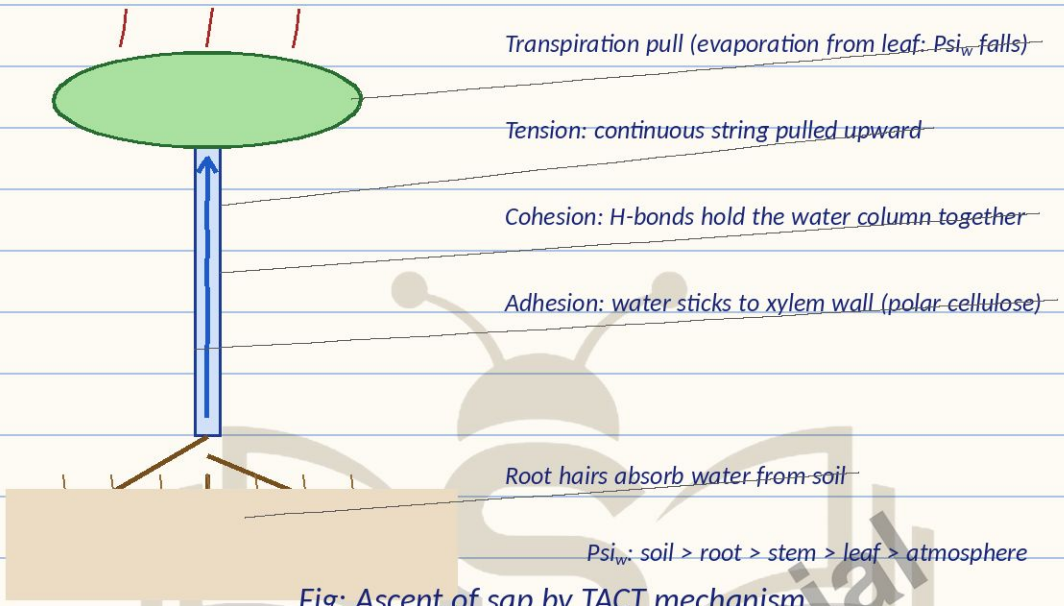


Fig: Ascent of sap by TACT mechanism

- *Transpiration pull from leaves; adhesion of water to xylem walls; cohesion between water molecules by hydrogen bonds; tension keeps the water column unbroken as it is pulled up, driven by solar energy.*

Q8. [SLO 9] Describe the mechanisms involved in opening and closing of stomata.

Ans:



Fig: Opening and closing of stomata

- *Starch-sugar theory: light, photosynthesis, glucose up, Psi_w down, water in,*

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turgid, open; dark: sugar to starch, Psi_w up, water out, flaccid, closed.

- *K^+ theory: light, low CO_2, K^+ influx + malate, potassium malate, water in, open; dark, high CO_2, K^+ efflux, water out, closed.*

Q9. [SLO 10] Explain the movement of sugars within plants.

Ans:

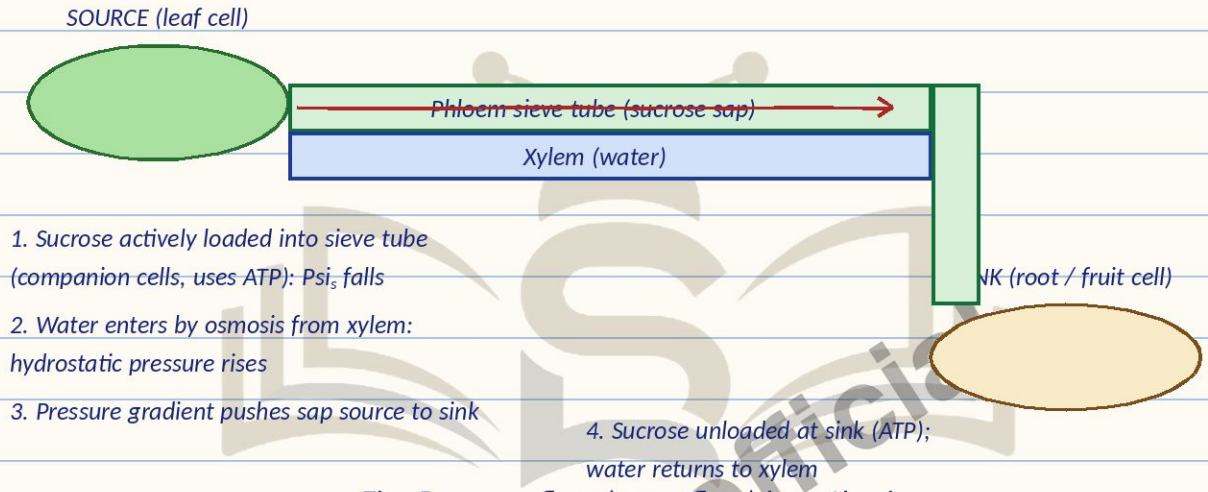


Fig: Pressure flow (mass flow) hypothesis

- *Translocation is from source to sink through phloem (sap 10-25% dry matter, 90% sucrose). Pressure flow: active loading at source, osmotic entry of water, high turgor pushes sap to sink, unloading at sink, water returns to xylem.*

Q10. [SLO 12] Explain movement of water into or out of the cell in isotonic, hypotonic and hypertonic conditions.

Ans:

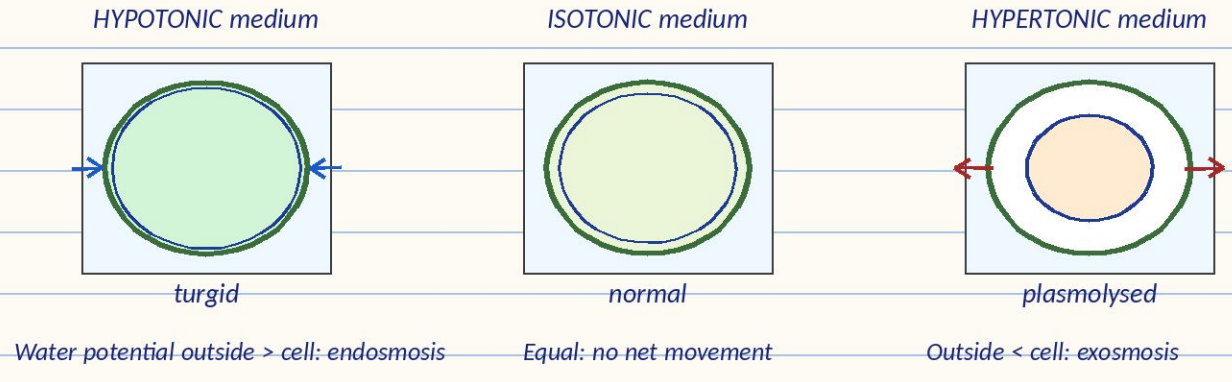


Fig: Cell in hypotonic, isotonic and hypertonic media

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- *Hypotonic: medium has higher Psi_w; endosmosis; turgid cell. Hypertonic: lower Psi_w; exosmosis; flaccid (plasmolysed). Isotonic: equal; no net movement.*

Q11. [SLO 13] Describe osmotic adjustments in hydrophytic, xerophytic and mesophytic plants.

Ans:

Feature	Hydrophytes	Mesophytes	Xerophytes
Medium	Hypotonic	Near isotonic	Hypertonic
Transpiration	Very high	Medium	Very low
Stomata	Upper, always open	Lower, open day	Sunken, open night
Cuticle / storage	Very thin / none	Thin / little	Thick / succulent

- *Marine and freshwater hydrophytes face flooding (hypotonic); desert xerophytes face dehydration.*

Q12. [SLO 13] Explain osmotic adjustment in plants of saline soils.

Ans:

- *Saline soil has very low Psi_w: physiological drought. Halophytes take up Na^+ and Cl^- and store them in leaf vacuoles, keeping cytoplasm salt low; the salt lowers cell Psi_w.*
- *Others do not take up salt but accumulate compatible solutes (glycine betaine, sorbitol) that do not harm enzymes.*

Q13. [SLO 14] List the adaptations in plants to cope with low and high temperature.

Ans:

- *Low: more unsaturated fatty acids in membranes; change of solute composition so cytosol supercools without ice crystals; gradual acclimation.*
- *High: shiny cuticle reflecting light; many stomata for evaporative cooling (close above 45 C); heat shock proteins protecting enzymes.*

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Q14. [SLO 15] Explain turgor pressure and its significance in providing support to herbaceous plants.

Ans:

- *Water fills the large central vacuoles of parenchyma, pressing on walls (turgor pressure) and making cells stiff. Herbs lack woody tissue, so turgor keeps them upright; fall in turgor causes wilting.*

Q15. [SLO 16] Describe the structure of supporting tissue in plants.

Ans:

Collenchyma	Sclerenchyma
Living, cellulose thickened corners	Dead, thick lignified walls
Young organs, herbs, petioles	Fibres (pericycle, xylem, phloem), sclereids (fruit wall, seed coat)
Can stretch and grow	Great tensile strength, rigid

- *Also xylem and phloem, ring of vascular bundles in dicot stem and secondary growth (wood).*

Q16. [SLO 17] Define growth and explain primary and secondary growth in plants.

Ans:

- *Growth: irreversible increase in size and mass (cell division, elongation, maturation, differentiation).*
- *Primary: lengthening by apical meristems (indeterminate). Secondary: thickening in dicots and gymnosperms by vascular cambium (secondary xylem inward, secondary phloem outward) and cork cambium (cork outward, secondary cortex inward).*

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Q17. [SLO 17] Describe the role of apical meristem and lateral meristem in primary and secondary growth.

Ans:

Apical	Lateral
Root and shoot tips	Sides of stem and root
Primary meristem; length	Secondary meristem; girth
Cells: initials and derivatives that differentiate	Vascular cambium, cork cambium, intercalary meristem

Q18. [SLO 18] Explain how annual rings are formed in plants.

Ans:

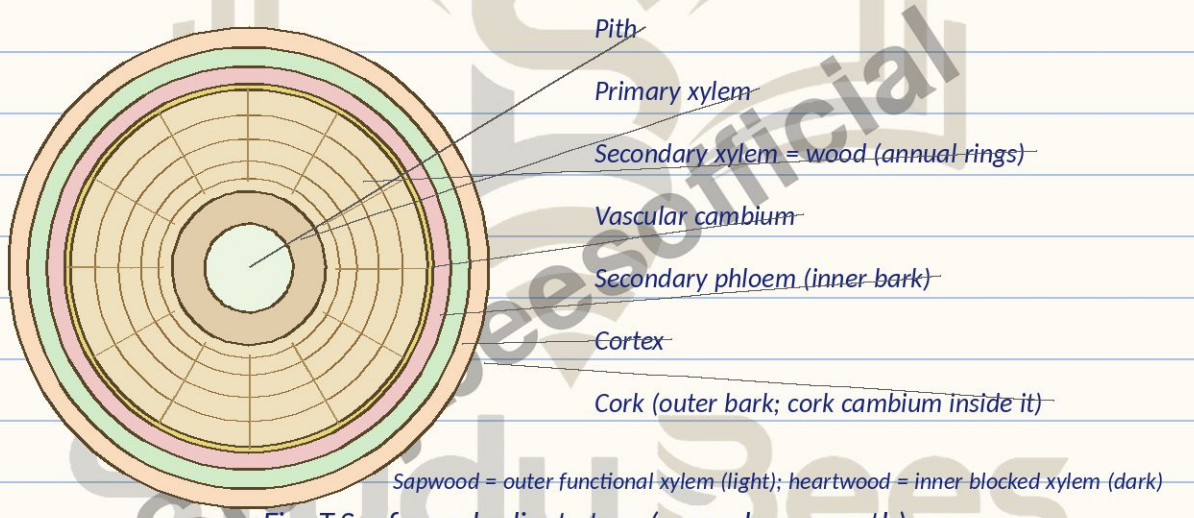


Fig: T.S. of woody dicot stem (secondary growth)

- Vascular cambium makes secondary xylem; in spring / summer (water available) large cells, in autumn / winter (water deficient) small cells. Alternating larger and smaller cells appear as one ring per year; counting rings gives the age.

Q19. [SLO 19] Explain the influence of apical meristem on the growth of lateral shoots.

Ans:

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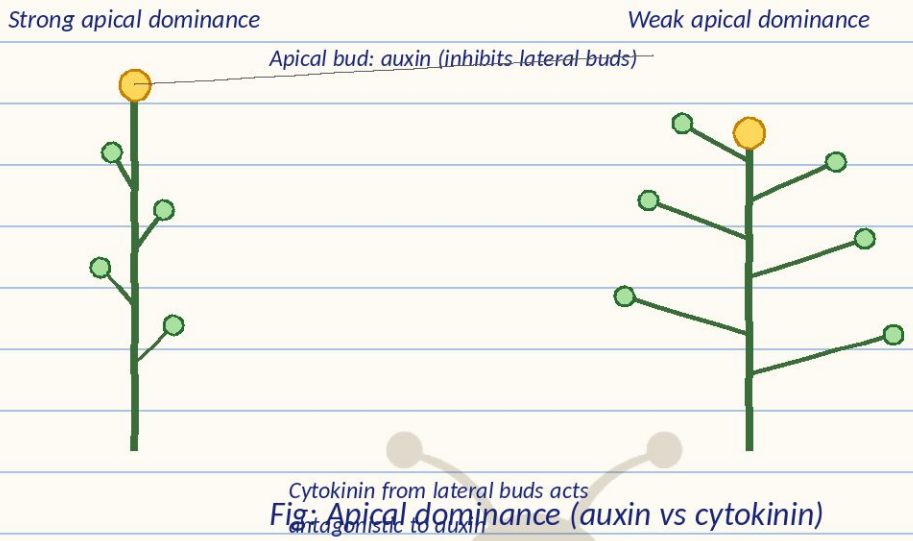


Fig: Apical dominance (auxin vs cytokinin)

- *Apical dominance: auxin from the apical bud inhibits lateral buds (negative correlation). Cytokinin from lateral buds is antagonistic. Strong dominance: tall plant with short branches; weak dominance: short bushy plant. Removal of the apical bud lets lateral buds grow.*

Q20. [SLO 20] Explain the role of important plant growth regulators.

Ans:

Regulator	***Role***
Auxin	*Cell elongation, apical dominance*
Gibberellin	*Stem elongation, seed germination, vernalin*
Cytokinin	*Cell division, lateral buds, delays senescence*
Abscisic acid	*Dormancy, stomatal closure, inhibits growth*
Ethene	*Fruit ripening, abscission*

Q21. [SLO 21] Explain types of movement in plants in response to light, gravity, touch and chemicals.

Ans:

- *Phototropism: shoots positive, roots negative. Geotropism: roots positive, shoots negative. Thigmotropism: climbers coil around support. Chemotropism: fungal hyphae toward nutrients.*
- *These tropisms are differential growth movements of organs towards or away*

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from stimuli, controlled by hormones.

Q22. [SLO 22] Classify plants on the basis of photoperiodism and give examples.

Ans:

Type	Examples
Short-day	Cocklebur, chrysanthemum, tobacco
Long-day	Spinach, sugar beet, henbane, snapdragon, cabbage, spring wheat
Day-neutral	Tomato, pansy, bean, sweet pea, rose

- Critical photoperiod: cocklebur 15.5 h, tobacco 14 h (SDP); henbane 11 h (LDP). The dark period is the real stimulus: SDP are long-night plants, LDP short-night plants.

Q23. [SLO 23] Explain the mechanism of photoperiodism with reference to the mode of action of phytochrome.

Ans:

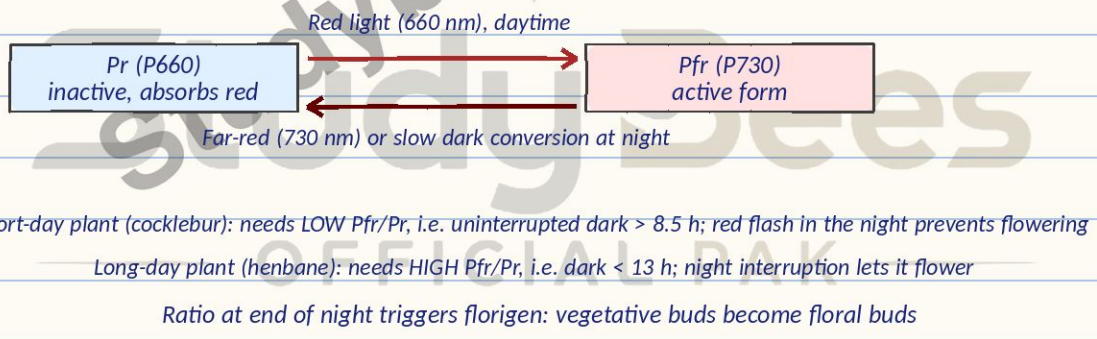


Fig: Phytochrome interconversion and photoperiodism

- Pr absorbs red (660 nm) in the day and changes to Pfr; Pfr changes back to Pr by far-red or slowly at night. The ratio at end of night measures night length (biological clock).
- SDP need low Pfr : Pr (dark more than 8.5 h); LDP need high Pfr : Pr (dark less than 13 h). The right ratio triggers florigen, which converts vegetative buds to floral buds.

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Q24. [SLO 24] Define vernalization and explain the role of low temperature treatment on flower production.

Ans:

- *Vernalization: promotion of flowering by low temperature (0-10 C, best about 4 C), received by the shoot apex.*
- *Biennials (winter wheat) need it: sown in autumn, they get cold in winter and flower next year; spring-sown seeds do not flower unless chilled. Low temperature stimulates vernalin (= gibberellin), which converts vegetative buds to floral buds.*

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